

ADVANCED

Stage 6

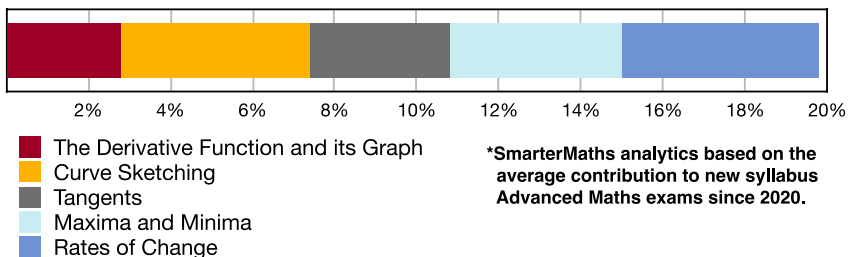
Calculus (Adv), C3 Applications of Calculus (Adv)

The Derivative Function and its Graph (Y12)

Teacher: Kristy Chen

Exam Equivalent Time: 84 minutes (based on allocation of 1.5 minutes per mark)

C3 Applications of Calculus



HISTORICAL CONTRIBUTION

- C3 Applications of Calculus* is the 800 pound gorilla in the Advanced rainforest, contributing a massive 19.8% to new syllabus exams since introduced in 2020.
- This topic has been split into five sub-topics for analysis purposes: *1-The Derivative Function and its Graph (2.8%), 2-Curve Sketching (4.6%), 3-Tangents (3.4%), 4-Maxima and Minima (4.2%) and 5-Rates of Change (4.8%)*.
- This analysis looks at *The Derivative Function and its Graph*.

HSC ANALYSIS - What to expect and common pitfalls

- The Derivative Function and its Graph* is a challenging sub-topic that has been examined in 4 out of 5 new syllabus exams (including two questions in 3 of those years).
- Students are regularly asked to explore the graphical relationship between $f(x)$ and $f'(x)$. This question type has caused problems in the past and deserves attention.
- Investigating graph shapes and the related values of $f'(x)$ and $f''(x)$ in certain domains is key revision that is well covered in the database (see *2024 Adv 11, 2021 Adv 7 MC*).
- The 2020 exam tested *composite functions* in this context for the first time, producing an 11% mean mark (lowest in the exam). This should be carefully reviewed (see *2020 Adv 10 MC*).

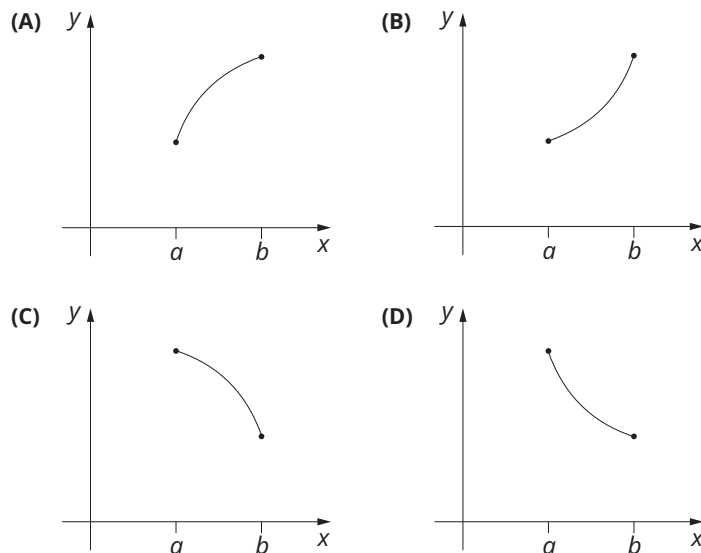
Questions

1. Calculus, 2ADV C3 2017 HSC 4 MC

The function $f(x)$ is defined for $a \leq x \leq b$.

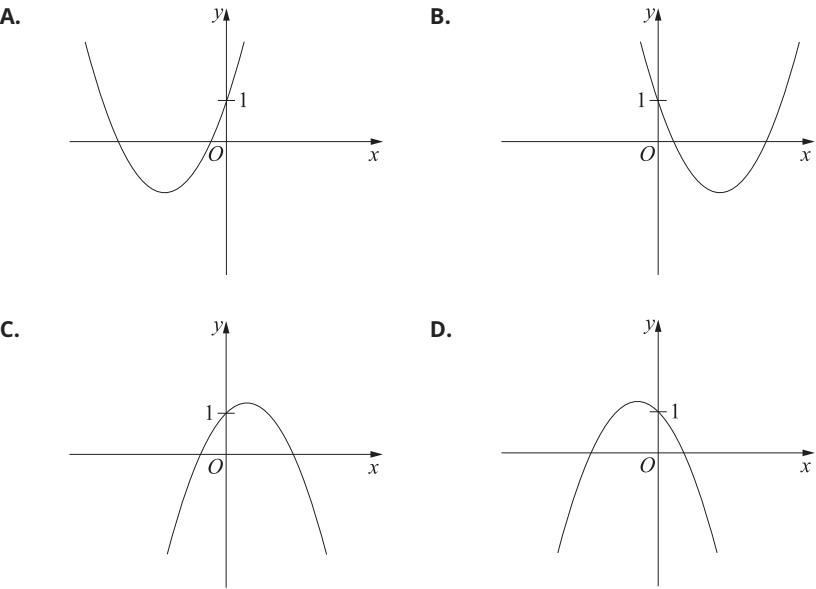
On this interval, $f'(x) > 0$ and $f''(x) < 0$.

Which graph best represents $y = f(x)$?



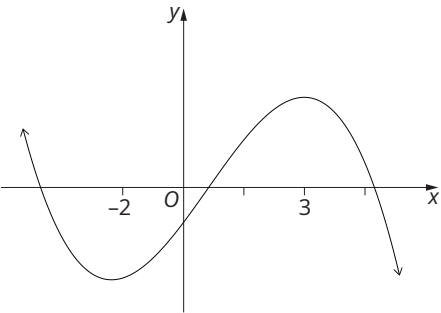
2. Calculus, 2ADV C3 2020 HSC 5 MC

Which of the following could represent the graph of $y = -x^2 + bx + 1$, where $b > 0$?



3. Calculus, 2ADV C3 2021 HSC 7 MC

The diagram shows part of $y = f(x)$ which has a local minimum at $x = -2$ and a local maximum at $x = 3$.



Which of the following shows the correct relationship between $f''(-2)$, $f(0)$ and $f'(3)$?

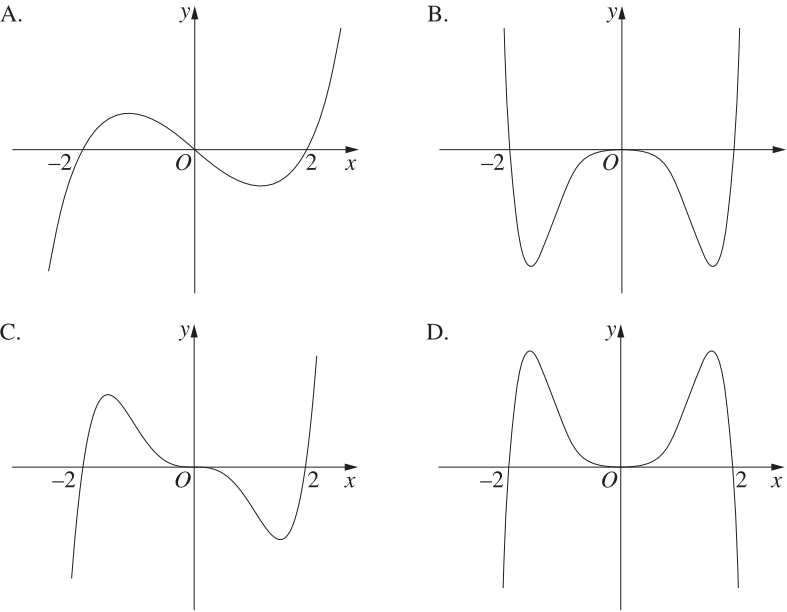
- A. $f(0) < f'(3) < f''(-2)$
- B. $f(0) < f''(-2) < f'(3)$
- C. $f''(-2) < f'(3) < f(0)$
- D. $f''(-2) < f(0) < f'(3)$

4. Calculus, 2ADV C3 2023 HSC 6 MC

The following table gives the signs of the first and second derivatives of a function $y = f(x)$ for different values of x .

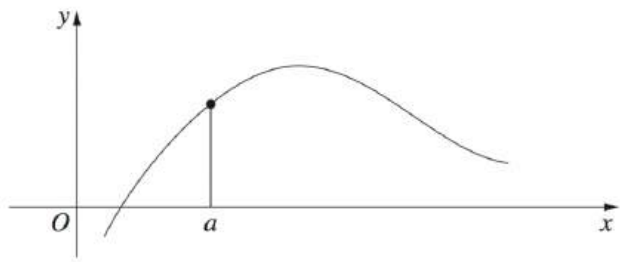
x	-2	0	2
$f'(x)$	$+$	0	$+$
$f''(x)$	$-$	0	$+$

Which of the following is a possible sketch of $y = f(x)$?



5. Calculus, 2ADV C3 2012 HSC 4 MC

The diagram shows the graph $f(x)$.



Which of the following statements is true?

- A. $f'(a) > 0$ and $f''(a) < 0$
- B. $f'(a) > 0$ and $f''(a) > 0$
- C. $f'(a) < 0$ and $f''(a) < 0$
- D. $f'(a) < 0$ and $f''(a) > 0$

6. Calculus, 2ADV C3 2024 13 MC

The function $f(x) = \frac{x}{2} + \frac{2}{x}$ undergoes the following sequence of transformations to become $g(x)$:

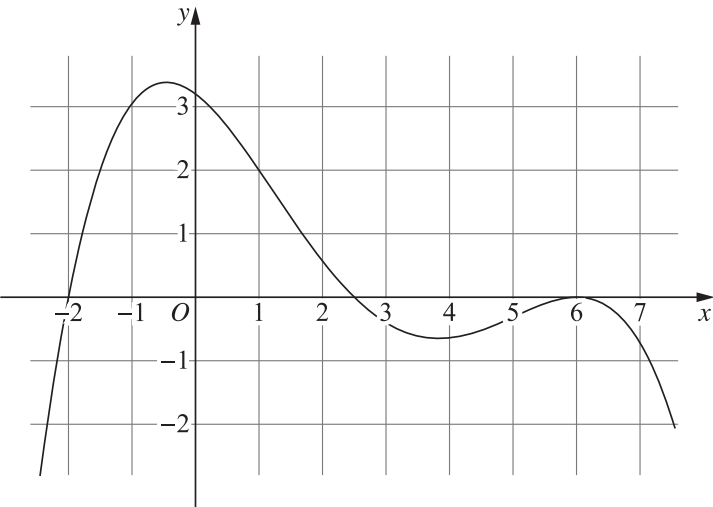
- 1. dilation by a factor of 3 from the y -axis
- 2. translation by 1 unit in the negative direction of the y -axis.

The function g has a local minimum at the point with the coordinates

- A. $(6, 1)$
- B. $\left(\frac{2}{3}, 1\right)$
- C. $(2, 5)$
- D. $\left(2, -\frac{1}{3}\right)$

7. Calculus, 2ADV C3 2025 HSC 9 MC

The diagram shows the graph of $y = f'(x)$.



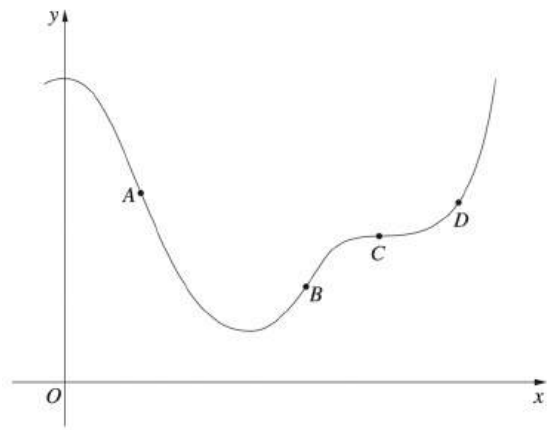
Given $f(1) = 6$, which interval includes the best estimate for $f(1.1)$?

- A. $[6.2, 6.4)$
- B. $[6.0, 6.2)$
- C. $[5.8, 6.0)$
- D. $[5.6, 5.8)$

8. Calculus, 2ADV C3 2013 HSC 8 MC

 *RAP Data (vs state mean) - Bottom 3%: School result (26%) was 22% below state average (48%)*

The diagram shows points A , B , C and D on the graph $y = f(x)$.

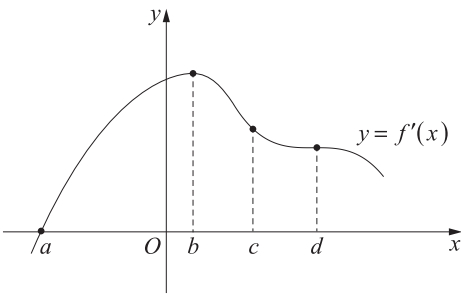


At which point is $f'(x) > 0$ and $f''(x) = 0$?

- A. A
- B. B
- C. C
- D. D

9. Calculus, 2ADV C3 2018 HSC 9 MC

The diagram shows the graph of $f'(x)$, the derivative of a function.

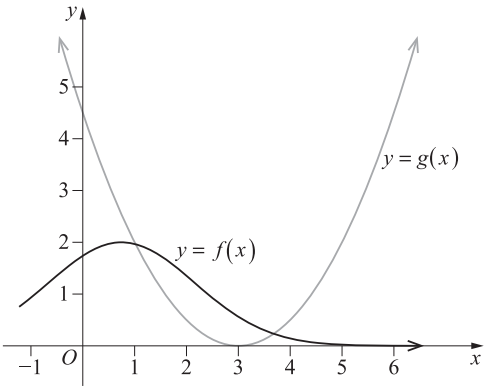


For what value of x does the graph of the function $f(x)$ have a point of inflection?

- A. $x = a$
- B. $x = b$
- C. $x = c$
- D. $x = d$

10. Calculus, 2ADV C3 2020 HSC 10 MC

The graph shows two functions $y = f(x)$ and $y = g(x)$.



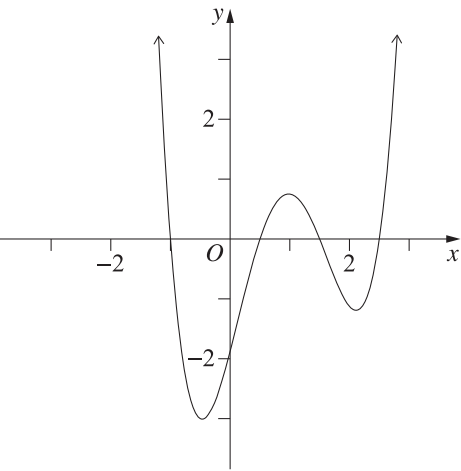
Define $h(x) = f(g(x))$.

How many stationary points does $y = h(x)$ have for $1 \leq x \leq 5$?

- A. 0
- B. 1
- C. 2
- D. 3

11. Calculus, 2ADV C3 2025 HSC 10 MC

The graph of $y = f(x)$, with all its stationary points, is shown.

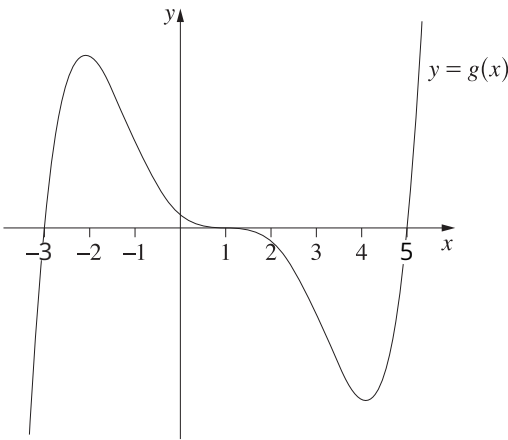


How many stationary points does the graph of $y = f(e^x)$ have?

- A. 0
- B. 1
- C. 2
- D. 3

12. Calculus, 2ADV C3 2024 HSC 11

The graph of the function $g(x)$ is shown.

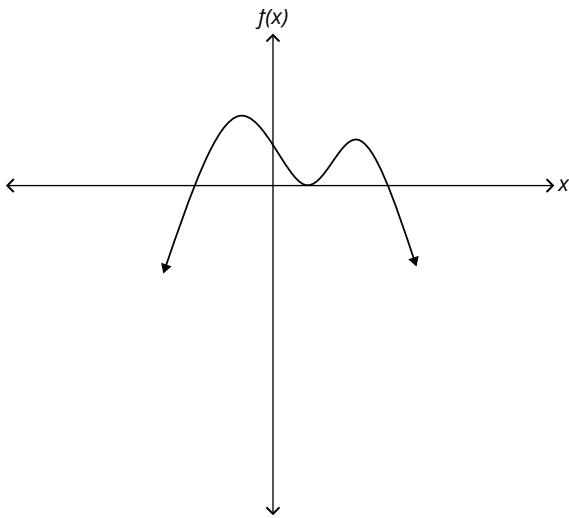


Using the graph, complete the table with the words positive, zero or negative as appropriate. (3 marks)

<i>x</i> -value	First derivative of $g(x)$ at x	Second derivative of $g(x)$ at x
$x = -3$		
$x = 1$		
$x = 5$		

13. Calculus, 2ADV C3 EQ-Bank 1

The graph of $f(x)$ is drawn below. Draw a sketch of $f'(x)$ on the same graph. (2 marks)



14. Calculus, 2ADV C3 2013 HSC 12a

The cubic $y = ax^3 + bx^2 + cx + d$ has a point of inflection at $x = p$.

Show that $p = -\frac{b}{3a}$. (2 marks)

15. Calculus, 2ADV C3 2021 HSC 16

For what values of x is $f(x) = x^2 - 2x^3$ increasing? (3 marks)

16. Calculus, 2ADV C3 2014 HSC 14a

 **RAP Data (vs state mean) - Bottom 17%: School result (60%) was 7% below state average (67%)**

Find the coordinates of the stationary point on the graph $y = e^x - ex$, and determine its nature. (3 marks)

17. Calculus, 2ADV C3 2024 MET2 3*

A function is defined as $f(x) = \frac{x - h}{(x + 1)(x - 4)}$

Determine the range of h where the graph of $f(x)$ have no turning points. (3 marks)

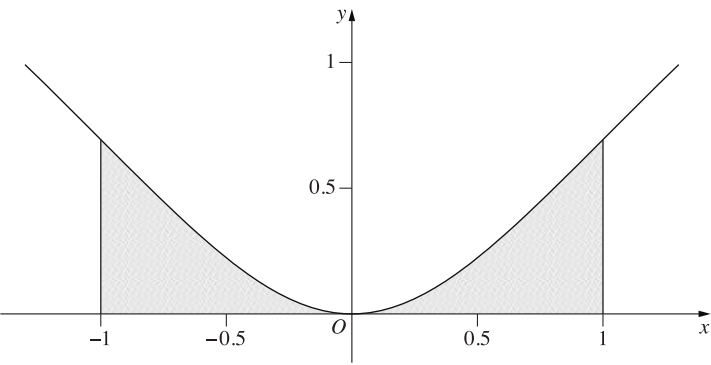
18. Calculus, 2ADV C3 SM-Bank 2

Given $y = xe^{-3x}$, prove that

$$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 0 \quad (3 \text{ marks})$$

19. Calculus, 2ADV C4 2024 HSC 22

The graph of the function $f(x) = \ln(1 + x^2)$ is shown.



a. Prove that $f(x)$ is concave up for $-1 < x < 1$. (3 marks)

b. A table of function values, correct to 4 decimal places, for some x values is provided.

x	0	0.25	0.5	0.75	1
$\ln(1 + x^2)$	0	0.0606	0.2231	0.4463	0.6931

Using the function values provided and the trapezoidal rule, estimate the shaded area in the diagram. (2 marks)

c. Is the answer to part (b) an overestimate or underestimate? Give a reason for your answer. (1 mark)

20. Calculus, 2ADV C3 2010 HSC 8d

Let $f(x) = x^3 - 3x^2 + kx + 8$, where k is a constant.

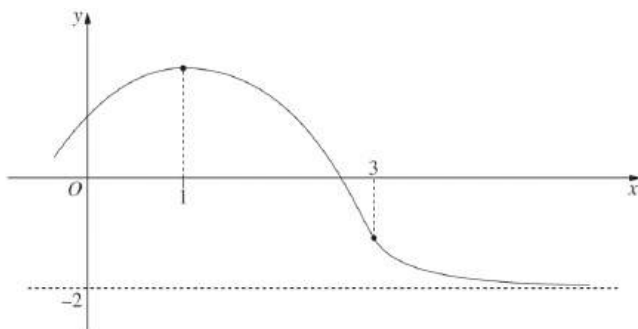
Find the values of k for which $f(x)$ is an increasing function. (2 marks)

21. Calculus, 2ADV C3 2011 HSC 9c

⚠ RAP Data (vs state mean) - Bottom 1%: School result (15%) was 28% below state average (43%)

⚠ RAP Data (raw) - School result was 15%

The graph $y = f(x)$ in the diagram has a stationary point when $x = 1$, a point of inflection when $x = 3$, and a horizontal asymptote $y = -2$.



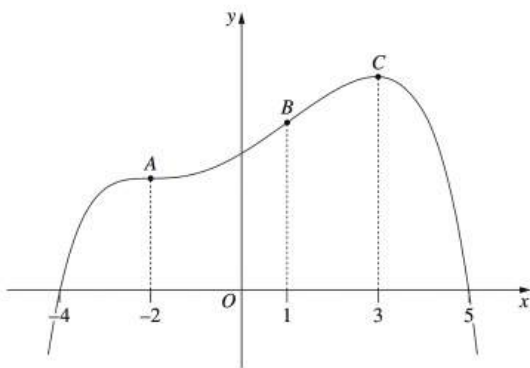
Sketch the graph $y = f'(x)$, clearly indicating its features at $x = 1$ and at $x = 3$, and the shape of the graph as $x \rightarrow \infty$. (3 marks)

22. Calculus, 2ADV C3 2014 HSC 14e

⚠ RAP Data (vs state mean) - Bottom 2%: School result (27%) was 24% below state average (51%)

The diagram shows the graph of a function $f(x)$.

The graph has a horizontal point of inflection at A , a point of inflection at B and a maximum turning point at C .



On the diagram above, sketch the graph of the derivative $f'(x)$. (3 marks)

23. Calculus, 2ADV C3 2018 HSC 14c

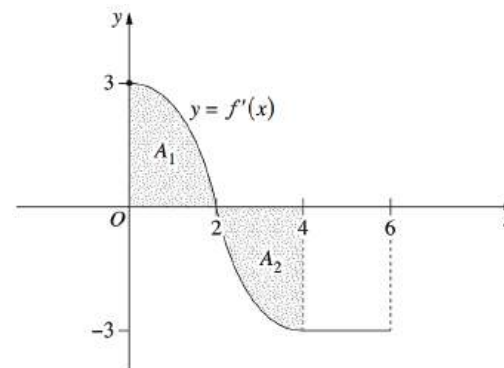
Let $f(x) = x^3 + kx^2 + 3x - 5$, where k is a constant.

Find the values of k for which $f(x)$ has NO stationary points. (3 marks)

24. Calculus, 2ADV C3 2010 HSC 9b

Let $y = f(x)$ be a function defined for $0 \leq x \leq 6$, with $f(0) = 0$.

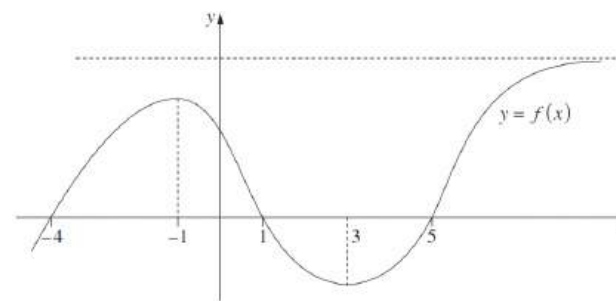
The diagram shows the graph of the derivative of f , $y = f'(x)$.



The shaded region A_1 has area 4 square units. The shaded region A_2 has area 4 square units.

- For which values of x is $f(x)$ increasing? (1 mark)
- What is the maximum value of $f(x)$? (1 mark)
- Find the value of $f(6)$. (1 mark)
- Draw a graph of $y = f(x)$ for $0 \leq x \leq 6$. (2 marks)

25. Calculus, 2ADV C3 2009 HSC 8a



The diagram shows the graph of a function $y = f(x)$.

- For which values of x is the derivative, $f'(x)$, negative? (1 mark)
- What happens to $f'(x)$ for large values of x ? (1 mark)
- Sketch the graph $y = f'(x)$. (2 marks)

Worked Solutions

1. Calculus, 2ADV C3 2017 HSC 4 MC

Interpreting $f'(x) > 0$,

$\Rightarrow$ gradient is always positive

Interpreting $f''(x) < 0$,

$\Rightarrow$ Curve is concave down

$\Rightarrow A$

2. Calculus, 2ADV C3 2020 HSC 5 MC

$$y = -x^2 + bx + 1$$

$$\frac{dy}{dx} = -2x + b$$

S.P. occurs when $\frac{dy}{dx} = 0$:

$$-2x + b = 0$$

$$x = \frac{b}{2}$$

Since $b > 0$, SP occurs when $x > 0$

$\Rightarrow C$

3. Calculus, 2ADV C3 2021 HSC 7 MC

$$f''(-2) > 0 \quad (\text{concave up at } x = -2)$$

$$f(0) < 0 \quad (\text{see graph})$$

$$f'(3) = 0 \quad (\text{S.P.})$$

$$\therefore f(0) < f'(3) < f''(-2)$$

$\Rightarrow A$

Mean mark 52%.

4. Calculus, 2ADV C3 2023 HSC 6 MC

By elimination:

Gradient is positive at $x = -2$ and 2 (Eliminate B and D)

SP and POI at $x = 0$ (Eliminate A)

$\Rightarrow C$

5. Calculus, 2ADV C3 2012 HSC 4 MC

At $x = a$,
 $f'(a) > 0 \quad \therefore$ Cannot be C or D
 $f''(a) < 0$ because $f(x)$ is concave down at $x = a$
 $\therefore$ Cannot be B
 $\Rightarrow A$

6. Calculus, 2ADV C3 2024 13 MC

Dilate by a factor of 3 from the y -axis:

$$f(x) \rightarrow f_1(x) = \frac{\frac{x}{3}}{2} + \frac{2}{\frac{x}{3}} = \frac{x}{6} + \frac{6}{x}$$

Translate 1 unit down:

$$f_1(x) \rightarrow g(x) = \frac{x}{6} + \frac{6}{x} - 1$$

$$g'(x) = \frac{1}{6} - \frac{6}{x^2}$$

Max/min when $\frac{1}{6} - \frac{6}{x^2} = 0$

$\Rightarrow A$

Mean mark 45%.

7. Calculus, 2ADV C3 2025 HSC 9 MC

Gradient of $f(x)$ at $x = 1$ is 2 (see graph).
Gradient of $f(x)$ at $x = 1.1$ is slightly below 2 (see graph).
As x increases 0.1 (from 1.0 to 1.1), y will increase less than 0.2 units.
 $\therefore f(1.1) \in [6.0, 6.2)$
 $\Rightarrow B$

8. Calculus, 2ADV C3 2013 HSC 8 MC

At A , $f'(x) < 0$
At C , $f'(x) = 0$
 $\therefore$ It cannot be A or C .
At D , $f''(x) > 0$ (concave up)
At B , $f''(x) = 0$ (concavity changes)
 $\Rightarrow B$

Mean mark 48%

9. Calculus, 2ADV C3 2018 HSC 9 MC

P.I. will occur when the graph of $f'(x)$
has a turning point.
 $\therefore x = b$
 $\Rightarrow B$

Mean mark 41%.

10. Calculus, 2ADV C3 2020 HSC 10 MC

$$h(x) = f(g(x))$$
$$h'(x) = g'(x) \times f'(g(x))$$

Mean mark 11%.

S.P.'s occur when $g'(x) = 0$ or $f'(g(x)) = 0$
 $g'(x) = 0$ when $x = 3$ (from graph)
 $f'(x) = 0$ when $x \approx 1$ (i.e. x -value is just under 1)

Find values of x when $g(x) \approx 1$:
By inspection, there are 2 values where
 $g(x) \approx 1$, $x \in [1, 5]$
 $\therefore$ There are 3 S.P.'s for $y = h(x)$, $x \in [1, 5]$
 $\Rightarrow D$

11. Calculus, 2ADV C3 2025 HSC 10 MC

$y = f(e^x) \Rightarrow y' = e^x \times f'(e^x)$
Find number of x values where $y' = 0$.
Since $e^x \in (0, \infty)$ for all x :
Stationary points of $f(e^x) = 2$ (SP's of $f(x)$ for $x \in (0, \infty)$).
 $\Rightarrow C$

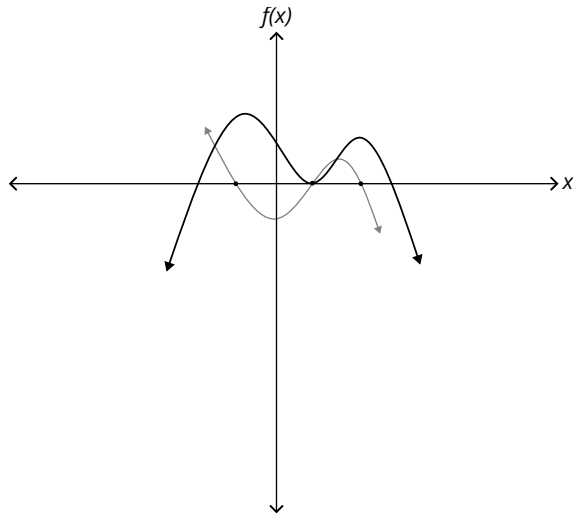
12. Calculus, 2ADV C3 2024 HSC 11

x -value	First derivative of $g(x)$ at x	Second derivative of $g(x)$ at x
$x = -3$	positive	negative
$x = 1$	zero	zero
$x = 5$	positive	positive

13. Calculus, 2ADV C3 EQ-Bank 1

Plot x-intercepts at the x-values of turning points.

Determine if gradient of $f(x)$ is positive/negative to draw shape.



14. Calculus, 2ADV C3 2013 HSC 12a

Show $p = -\frac{b}{3a}$

$$y = ax^3 + bx^2 + cx + d$$

$$y' = 3ax^2 + 2bx + c$$

$$y'' = 6ax + 2b$$

Given P.I. occurs when $x = p$

$$\Rightarrow y'' = 0 \text{ when } x = p$$

$$\therefore 6ap + 2b = 0$$

$$6ap = -2b$$

$$p = -\frac{2b}{6a}$$

$$= -\frac{b}{3a} \quad \dots \text{ as required}$$

15. Calculus, 2ADV C3 2021 HSC 16

$$f(x) = x^2 - 2x^3$$

$$f'(x) = 2x - 6x^2$$

$f(x)$ is increasing when $f'(x) > 0$

$$2x - 6x^2 > 0$$

$$2x(1 - 3x) > 0$$

$$x \in \left(0, \frac{1}{3}\right)$$

16. Calculus, 2ADV C3 2014 HSC 14a

$$y = e^x - ex$$

$$\frac{dy}{dx} = e^x - e$$

$$\frac{d^2y}{dx^2} = e^x$$

S.P. when $\frac{dy}{dx} = 0$

$$e^x - e = 0$$

$$e^x = e^1$$

$$x = 1$$

At $x = 1$

$$y = e^1 - e = 0$$

$$\frac{d^2y}{dx^2} = e > 0 \Rightarrow \text{MIN}$$

$\therefore$ MINIMUM S.P. at $(1, 0)$

17. Calculus, 2ADV C3 2024 MET2 3*

$$f(x) = \frac{x - h}{(x + 1)(x - 4)}$$

Let $u = x - h \Rightarrow u' = 1$

$$v = x^2 - 3x - 4 \Rightarrow v' = 2x - 3$$

$$f'(x) = \frac{(x^2 - 3x - 4) - (2x - 3)(x - h)}{(x + 1)^2(x - 4)^2}$$

Solve $f'(x) = 0$:

$$x^2 - 3x - 4 - 2x^2 + 2xh + 3x - 3h = 0$$

$$-x^2 + 2xh - (4 + 3h) = 0$$

$$x^2 - 2xh + (4 + 3h) = 0$$

$$x = \frac{2h \pm \sqrt{4h^2 - 4(4 + 3h)}}{2}$$

$$= h \pm \sqrt{h^2 - 3h - 4}$$

No turning points occur if $h^2 - 3h - 4 = (h - 4)(h + 1) < 0$

$$-1 < h < 4$$

If $h = -1$ or 4 , $f(x)$ is linear (no TPs)

$$\therefore -1 \leq h \leq 4$$

18. Calculus, 2ADV C3 SM-Bank 2

$$y = xe^{-3x}$$

$$\frac{dy}{dx} = e^{-3x} - 3xe^{-3x}$$

$$\frac{d^2y}{dx^2} = -3e^{-3x} - 3e^{-3x} + 3 \cdot 3xe^{-3x}$$

$$= -6e^{-3x} + 9xe^{-3x}$$

Substituting into equation:

$$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y$$

$$= -6e^{-3x} + 9xe^{-3x} + 6(e^{-3x} - 3xe^{-3x}) + 9xe^{-3x}$$

$$= -6e^{-3x} + 9xe^{-3x} + 6e^{-3x} - 18xe^{-3x} + 9xe^{-3x}$$

$$= 0$$

19. Calculus, 2ADV C4 2024 HSC 22

a. $f(x) = \ln(1 + x^2)$

Mean mark (a) 47%.

$$f'(x) = \frac{2x}{1 + x^2}$$

$$f''(x) = \frac{2(1 + x^2) - 2x(2x)}{(1 + x^2)^2}$$

$$= \frac{2 + 2x^2 - 4x^2}{(1 + x^2)^2}$$

$$= \frac{2(1 - x^2)}{(1 + x^2)^2}$$

In domain $x \in (-1, 1) \Rightarrow 1 - x^2 > 0$

$$(1 + x^2)^2 > 0 \text{ for all } x$$

$$\Rightarrow f''(x) > 0 \text{ for } x \in (-1, 1)$$

$$\therefore f(x) \text{ is concave up for } x \in (-1, 1)$$

b. Total shaded area

Mean mark (b) 50%.

$$\approx 2 \times \frac{h}{2} [y_0 + 2(y_1 + y_2 + y_3) + y_4]$$

$$\approx 2 \times \frac{0.25}{2} [0 + 2(0.0606 + 0.2231 + 0.4463) + 0.6931]$$

$$\approx 0.538275$$

$$\approx 0.5383 \text{ (4 d.p.)}$$

c. Trapezoidal rule assumes straight lines join points in the table.

Mean mark (c) 49%.

Since the graph is concave up in the given domain, the trapezoidal rule will overestimate the area.

20. Calculus, 2ADV C3 2010 HSC 8d

$$f(x) = x^3 - 3x^2 + kx + 8$$

$$f'(x) = 3x^2 - 6x + k$$

$f(x)$ is increasing when $f'(x) > 0$

$$\Rightarrow 3x^2 - 6x + k > 0$$

$f'(x)$ is always positive

$\Rightarrow f'(x)$ is a positive definite.

i.e. when $a > 0$ and $\Delta < 0$

$$a = 3 > 0$$

$$\Delta = b^2 - 4ac$$

$$\therefore (-6)^2 - (4 \times 3 \times k) < 0$$

$$36 - 12k < 0$$

$$12k > 36$$

$$k > 3$$

$\therefore f(x)$ is increasing when $k > 3$.

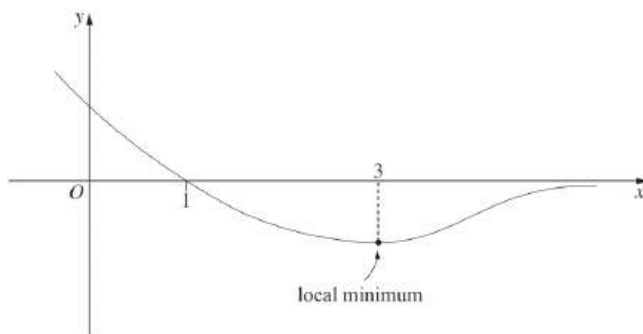
Mean mark 28%.

MARKER'S COMMENT: The arithmetic required to solve $36 - 12k < 0$ proved the undoing of *many* students.

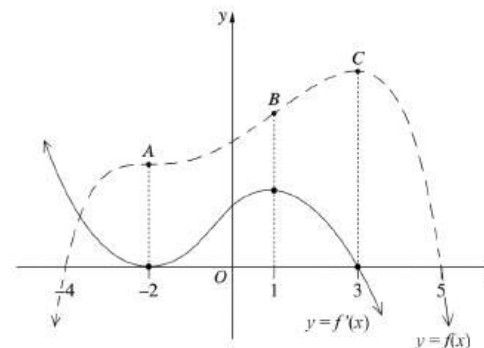
21. Calculus, 2ADV C3 2011 HSC 9c

Mean mark 43%

IMPORTANT: Examiners *regularly* ask questions that require the graphing of an $f'(x)$ given the $f(x)$ graph and vice-versa. *KNOW IT!*



22. Calculus, 2ADV C3 2014 HSC 14e



23. Calculus, 2ADV C3 2018 HSC 14c

$$f(x) = x^3 + kx^2 + 3x - 5$$

$$f'(x) = 3x^2 + 2kx + 3$$

Mean mark 49%.

No S.P.'s exist if $f'(x)$ has no roots,

$$\Delta < 0$$

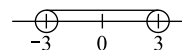
$$b^2 - 4ac < 0$$

$$(2k)^2 - 4 \times 3 \times 3 < 0$$

$$4k^2 - 36 < 0$$

$$k^2 - 9 < 0$$

$$(k-3)(k+3) < 0$$



$$\therefore -3 < k < 3$$

24. Calculus, 2ADV C3 2010 HSC 9b

- i. $f(x)$ is increasing when $f'(x) > 0$

From the graph

$f(x)$ is increasing when $0 \leq x < 2$

- ii. $f'(x) = 0$ when $x = 2$

$\therefore$ MAX at $x = 2$

$$\int_0^2 f'(x) dx = 4 \quad (\text{given since } A_1 = 4)$$

We also know

$$\begin{aligned} \int_0^2 f'(x) dx &= [f(x)]_0^2 \\ &= f(2) - f(0) \\ &= f(2) \quad (\text{since } f(0) = 0) \\ \Rightarrow f(2) &= 4 \end{aligned}$$

$\therefore$ MAX value of $f(x) = 4$

iii. $\int_0^4 f'(x) dx = A_1 - A_2$

$$= 0$$

We also know

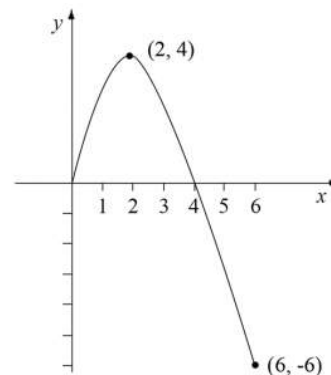
$$\begin{aligned} \int_0^4 f'(x) dx &= \int_2^4 f'(x) dx + \int_0^2 f'(x) dx \\ &= [f(x)]_2^4 + 4 \\ &= f(4) - f(2) + 4 \quad (\text{note } f(2) = 4) \\ &= f(4) \\ \Rightarrow f(4) &= 0 \end{aligned}$$

Gradient $= -3$ from $x = 4$ to $x = 6$

$$\begin{aligned} \therefore f(6) &= -3(6 - 4) \\ &= -6 \end{aligned}$$

Parts (ii) and (iii) proved particularly difficult for students with mean marks of 12% and 11% respectively.

iv.



Mean mark 28%
EXAM TIP: Clearly identify *THE EXTREMES* when given a defined domain. In this case, the origin is obvious graphically, and the other extreme at $x = 6$, is *CLEARLY LABELLED!*

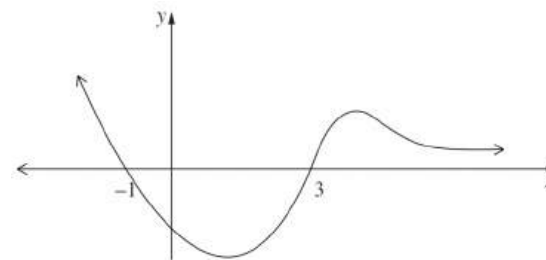
25. Calculus, 2ADV C3 2009 HSC 8a

- i. $f'(x) < 0$ when
 $-1 < x < 3$

- ii. As $x \rightarrow \infty$,
 $f'(x) \rightarrow 0$

Exact data not available.

iii.



Exact data not available.
MARKER'S COMMENT: Poorly drawn graphs with axes not labelled and inaccurate scales were common.